Sicher C Modules

Start

Tokeniser
(C Source File to '*Valid Sicher C Tokens*')

AST
|
IR

Parser
(Checks the eligibility of the C Source File compared to 'legal Sicher C Code')
(Parses a valid Sicher C Node to the AST which can then be turned into IR and optimised)

Semantic Analysis
(Type Checking)
(Declaration Checking)
(Compile Time Optimisations)

Code Generation
Generates the IR Sicher Representation to 'Machine Code'
(Hopefully targeting x64 ELF Linux Binaries)

Visualiser
Output to a 2D GUI, the in's and outs of Sicher C and all of it's modern enhancements and any memory safety mitigations applied

End

Sicher C Sub-Modules

- Unutilised Memory Object Removal
- Constant Folding
- CSE
- Basic Branch Prediction
- Basic Pipelining